

Financial Modelling Homework – Bence Dáné

Part 1: Visa vs Mastercard

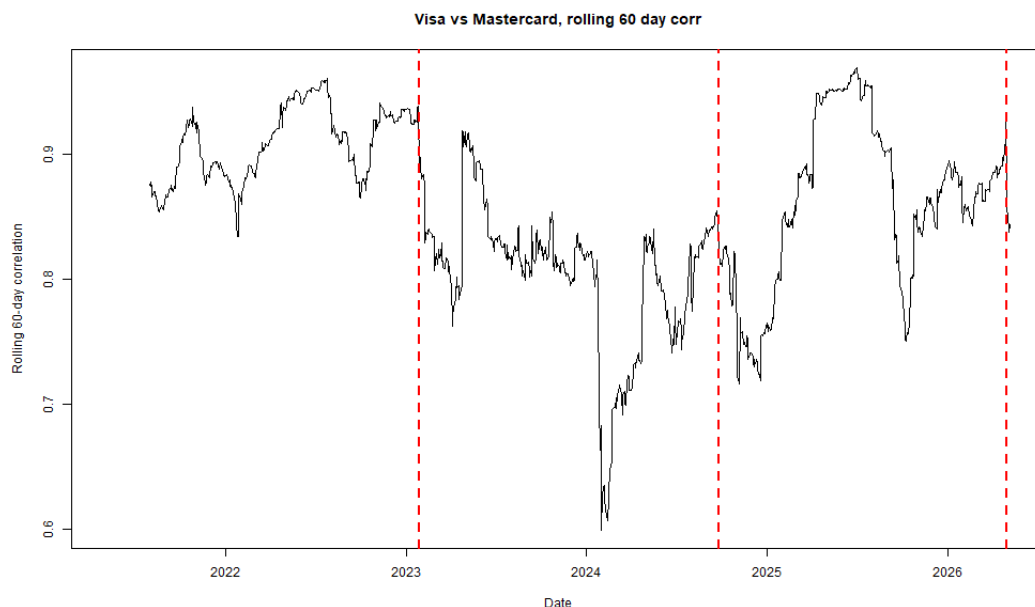
Date	Visa	Mastercard	V - MA	News
2026-04-29	8.26%	3.47%	4.79%	Footnote: 1
2023-01-27	2.99%	-0.85%	3.85%	Footnote: 2, 3
2024-09-24	-5.49%	-1.96%	-3.53%	Footnote: 4

Explanation

2026-04-29: Visa rose **8.26%** versus Mastercard's **3.47%**. Visa's quarter ended on March 31, with a jump of 17% over a year, landing at \$11.2 billion.¹

2023-01-27: Visa rose **2.99%**, while Mastercard fell **0.85%**. Visa reported Q1 2023 on January 26; revenue was up by 12%. Mastercard's revenue was below Wall Street expectations.^{2 3}

2024-09-24: Visa fell **5.49%** versus Mastercard's **1.96%** decline. The reason behind Visa's performance was Justice Department suing Visa for monopolizing debit markets. Since Mastercard is in the same sector, its decline is related to Visa's.⁴



¹ <https://www.fool.com/investing/2026/04/29/why-visa-stock-is-up-today/>

² <https://investor.visa.com/news/news-details/2023/Visa-Inc.-Fiscal-First-Quarter-2023-Financial-Results/default.aspx>

³ <https://www.investing.com/news/stock-market-news/mastercard-shares-fall-as-credit-card-giant-q1-revenue-outlook-misses-estimates-2989068>

⁴ <https://www.justice.gov/archives/opa/pr/justice-department-sues-visa-monopolizing-debit-markets>

Correlation

The 60-day correlation is $\text{Corr}(V, MA) = \text{Cov}(V, MA) / (\text{SD}(V) * \text{SD}(MA))$.

Date	Corr with	Corr without	Cov with	Cov without	SD V with	SD V without	SD MA with	SD MA without
2026-04-29	0.870	0.925	0.000224	0.000176	1.68%	1.29%	1.53%	1.47%
2023-01-27	0.891	0.933	0.000278	0.000288	1.66%	1.63%	1.88%	1.89%
2024-09-24	0.821	0.851	0.000141	0.000122	1.37%	1.17%	1.25%	1.23%

All outlier days reduced the 60-day correlation. Because of the above-mentioned lawsuit Visa's SD jumped from 1.17% to 1.37%, while Mastercard's barely changed.

Part 2: Rolling Beta

For the 10-year beta analysis, NVIDIA and SP500 were chosen. Rolling beta was computed over 252 days, where $\text{Beta} = \text{Cov}(NVDA, SP500) / \text{Var}(SP500)$.

Measure	Before	After	Change
Correlation rho	0.728	0.823	0.094
NVDA daily SD	2.27%	3.51%	1.23%
SP500 daily SD	0.74%	1.96%	1.22%
Volatility ratio	3.055	1.785	-1.270
Beta	2.225	1.469	-0.756



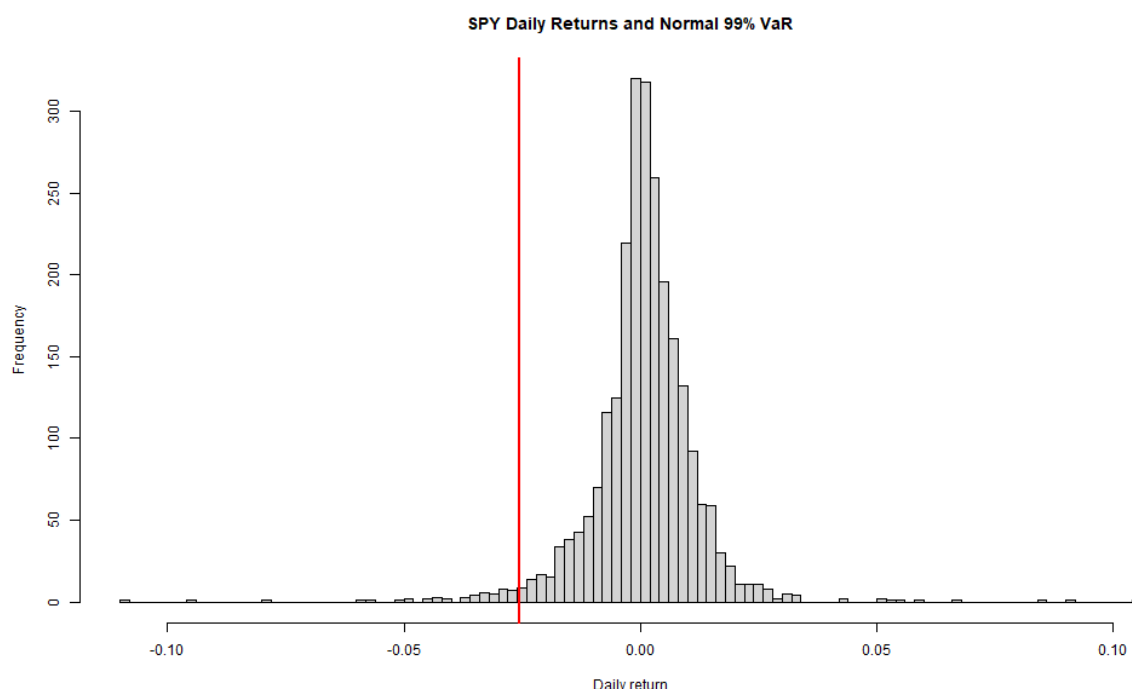
The regime's starting and ending point was identified using the following chart (the red lines represent the chosen period between **2020-01-28** and **2020-04-23**). That period lines up with the COVID-19 crash.⁵

Beta fell even though correlation rose. The main driver was the volatility ratio: market volatility increased sharply during the COVID crash, so NVIDIA's volatility relative to the market fell from 3.081 to 1.732.

Part 3: Normal VaR

SPY's normal 99% daily VaR was calculated using the mean and standard deviation of daily returns over the 10-year window.

Measure	Value
Index	SPY
Start date	2016-05-09
End date	2026-05-05
Observations	2512
Normal 99% daily VaR	2.57%
Actual breach count	49
Actual breach rate	1.95%



⁵ <https://www.federalreserve.gov/econres/notes/feds-notes/banks-backtesting-exceptions-during-the-covid-19-crash-causes-and-consequences-accessible-20210708.htm>